



Preview of this week

- In week 1, you learned that a particle filter seeks to evaluate:

$$f(x_k | \mathbb{Z}_{k-1}) = \int_{R_{x_{k-1}}} f(x_k | x_{k-1}) f(x_{k-1} | \mathbb{Z}_{k-1}) dx_{k-1}$$

$$f(x_k | \mathbb{Z}_k) = \frac{f(z_k | x_k) f(x_k | \mathbb{Z}_{k-1})}{f(z_k | \mathbb{Z}_{k-1})}.$$

- Last week, you learned how to approximate the integrals in these expressions using efficient Monte–Carlo methods.
 - This involved learning about importance densities, weight normalization, the impulse function, and how to visualize a pdf stored as a sum of impulses.
- This week, we combine all of these tools together to develop and refine the particle-filter algorithm.
- In this lesson, at last, we develop our first particle-filter algorithm!



Clarifying the objective of the particle filter

- Our goal is to estimate either the joint or marginal pdf:

$$f(\mathbb{X}_k | \mathbb{Z}_k) \quad \text{or} \quad f(x_k | \mathbb{Z}_k).$$

- The first of these is the pdf of the entire state trajectory $\mathbb{X}_k = \{x_0, x_1, \dots, x_k\}$ up until time k given all measurements up until time k .
 - That is, what is the likelihood of any state trajectory, given the measurements?
 - This pdf has high (and growing) dimension ($\mathbb{X}_k$ has high and growing dimension).
- The second of these is the pdf of only the present state vector x_k given all measurements up until time k .
 - That is, what is the likelihood that the present state vector takes on different values, given the measurements?
 - This pdf holds less information but has constant dimension over time.
- We will work with the joint pdf for now and simplify to the marginal pdf later.



Key idea

- We approximate the joint pdf $f(\mathbb{X}_k | \mathbb{Z}_k)$ as:

$$\hat{f}(\mathbb{X}_k | \mathbb{Z}_k) = \sum_{i=1}^{N_p} w_k^{(i)} \delta(\mathbb{X}_k - \mathbb{X}_k^{(i)}), \quad \sum_{i=1}^{N_p} w_k^{(i)} = 1.$$

- This discrete weighted approximation is completely parameterized by $\{\mathbb{X}_k^{(i)}, w_k^{(i)}\}_{i=1}^{N_p}$.
- We find these parameters using a Monte-Carlo approach to evaluating the particle-filter integrals. So,
 - We will need to select an importance density for the prediction step because we cannot sample from the pdfs involved in the integrals.
 - We will also need a recursive weight-update step.



Design requirements on the importance density

- Any pdf (including an importance density) must factor:

$$q(\mathbb{X}_k | \mathbb{Z}_k) = q(x_k | \mathbb{X}_{k-1}, \mathbb{Z}_k) q(\mathbb{X}_{k-1} | \mathbb{Z}_k).$$

- We *choose* an importance density that factors as follows:

$$q(\mathbb{X}_k | \mathbb{Z}_k) = q(x_k | \mathbb{X}_{k-1}, \mathbb{Z}_k) q(\mathbb{X}_{k-1} | \mathbb{Z}_{k-1}).$$

- Still not quite recursive, so again, we *choose* an importance density that satisfies:

$$q(\mathbb{X}_k | \mathbb{Z}_k) = q(x_k | x_{k-1}, z_k) q(\mathbb{X}_{k-1} | \mathbb{Z}_{k-1}).$$

- With this factoring, samples of the trajectory $\mathbb{X}_k^{(i)}$ can be created recursively after measuring z_k by appending $x_k^{(i)}$ to $\mathbb{X}_{k-1}^{(i)}$, where $x_k^{(i)}$ is made from z_k and $x_{k-1}^{(i)}$ only.

- Note: $x_k^{(i)}$ is the k th member of “particle” $\mathbb{X}_k^{(i)}$, where the particle comprises the entire trajectory and $x_k^{(i)}$ is only the most recent point in trajectory;
- Similarly, $x_{k-1}^{(i)}$ is the penultimate point in $\mathbb{X}_k^{(i)}$, or the final point in $\mathbb{X}_{k-1}^{(i)}$.



Updating particle weights

- With the requirements that we have placed on the importance density $q(\mathbb{X}_k | \mathbb{Z}_k)$, you can now update $\mathbb{X}_k^{(i)}$ recursively in:

$$\hat{f}(\mathbb{X}_k | \mathbb{Z}_k) = \sum_{i=1}^{N_p} w_k^{(i)} \delta(\mathbb{X}_k - \mathbb{X}_k^{(i)}), \quad \sum_{i=1}^{N_p} w_k^{(i)} = 1.$$

- We still need a way to evaluate the weight-update equation:

$$w_k^{(i)} \propto \frac{f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)}{q(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)}.$$

- Since we design $q(\mathbb{X}_k | \mathbb{Z}_k)$, we can compute the denominator term directly.
- It remains to determine how to calculate $f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)$ recursively.



Calculating $f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)$ recursively

- Note that we can write:

$$\begin{aligned} f(\mathbb{X}_k^{(i)}, z_k | \mathbb{Z}_{k-1}) &= f(\mathbb{X}_k^{(i)} | z_k, \mathbb{Z}_{k-1}) f(z_k | \mathbb{Z}_{k-1}) \\ &= f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k) f(z_k | \mathbb{Z}_{k-1}). \end{aligned}$$

- We can also write:

$$\begin{aligned} f(\mathbb{X}_k^{(i)}, z_k | \mathbb{Z}_{k-1}) &= f(z_k | \mathbb{Z}_{k-1}, \mathbb{X}_k^{(i)}) f(\mathbb{X}_k^{(i)} | \mathbb{Z}_{k-1}) \\ &= f(z_k | x_k^{(i)}) f(\mathbb{X}_k^{(i)} | \mathbb{Z}_{k-1}) \\ &= f(z_k | x_k^{(i)}) f(x_k^{(i)}, \mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1}) \\ &= f(z_k | x_k^{(i)}) f(x_k^{(i)} | \mathbb{Z}_{k-1}, \mathbb{X}_{k-1}^{(i)}) f(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1}) \\ &= f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)}) f(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1}). \end{aligned}$$

- We will equate these two forms to find $f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)$.



Computing the weight update

- Equating these two forms gives:

$$\begin{aligned} f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k) &= \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{f(z_k | \mathbb{Z}_{k-1})} f(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1}) \\ &\propto f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)}) f(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1}). \end{aligned}$$

- Recall that: $q(\mathbb{X}_k^{(i)} | \mathbb{Z}_k) = q(x_k^{(i)} | x_{k-1}^{(i)}, z_k) q(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1})$.
- So, we conclude that the weight update is proportional to:

$$\begin{aligned} w_k^{(i)} &\propto \frac{f(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)}{q(\mathbb{X}_k^{(i)} | \mathbb{Z}_k)} \propto \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} \frac{f(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1})}{q(\mathbb{X}_{k-1}^{(i)} | \mathbb{Z}_{k-1})} \\ &= \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} w_{k-1}^{(i)}. \end{aligned}$$



Perspective

- We now have a clever way to estimate the joint pdf:

$$\hat{f}(\mathbb{X}_k^{(i)} | \mathbb{Z}_k) = \sum_{i=1}^{N_p} w_k^{(i)} \delta(\mathbb{X}_k - \mathbb{X}_k^{(i)}).$$

- We get to choose $q(\cdot)$, from which we draw $x_k^{(i)}$ to append to $\mathbb{X}_{k-1}^{(i)}$ to form $\mathbb{X}_k^{(i)}$.
- This depends only on $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$, which depends on z_k and $x_{k-1}^{(i)}$.
- We then update the unnormalized weights recursively:

$$\tilde{w}_k^{(i)} = \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} \tilde{w}_{k-1}^{(i)}.$$

- $f(z_k | x_k^{(i)})$ is obtained from the measurement model $z_k = h_k(x_k, u_k, v_k)$.
- $f(x_k^{(i)} | x_{k-1}^{(i)})$ is from the process model $x_k = f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1})$.



What if we want the marginal pdf instead?

- But, what if we want only the marginal pdf $f(x_k^{(i)} | \mathbb{Z}_k)$?
- We can find this by integrating the joint pdf:

$$\hat{f}(x_k | \mathbb{Z}_k) = \int_{R_{\mathbb{X}_{k-1}}} \hat{f}(\mathbb{X}_k | \mathbb{Z}_k) d\mathbb{X}_{k-1} = \int_{R_{\mathbb{X}_{k-1}}} \sum_{i=1}^{N_p} w_k^{(i)} \delta(\mathbb{X}_k - \mathbb{X}_k^{(i)}) d\mathbb{X}_{k-1},$$

where $\delta(\mathbb{X}_k) = \prod_{i=0}^n \delta(x_i) = \delta(x_0) \delta(x_1) \cdots \delta(x_n)$.

- So, taking advantage of the fact that the integral of an impulse over its range is one,

$$\hat{f}(x_k | \mathbb{Z}_k) = \sum_{i=1}^{N_p} w_k^{(i)} \int_{R_{\mathbb{X}_{k-1}}} \delta(x_0 - x_0^{(i)}) \cdots \delta(x_k - x_k^{(i)}) d\mathbb{X}_{k-1} = \sum_{i=1}^{N_p} w_k^{(i)} \delta(x_k - x_k^{(i)}).$$

- The impulse notation makes the algorithm completely recursive if only the marginal is needed.



Sequential Importance Sampling (SIS) algorithm

- So, the most basic general particle-filter algorithm, known as “sequential importance sampling,” can be written as:

$$\{x_k^{(i)}, w_k^{(i)}\}_{i=1}^{N_p} = \text{SIS} \left(\{x_{k-1}^{(i)}, w_{k-1}^{(i)}\}_{i=1}^{N_p}, z_k \right)$$

For $i = 1 : N_p$,

Draw new particle value $x_k^{(i)} \sim q(x_k | x_{k-1}^{(i)}, z_k)$ — randomly!

Calculate unnormalized weight for this particle:

$$\tilde{w}_k^{(i)} = \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} w_{k-1}^{(i)}.$$

End for loop

Normalize the weights for all particles: $w_k^{(i)} = \frac{\tilde{w}_k^{(i)}}{\sum_{i=1}^{N_p} \tilde{w}_k^{(i)}}.$



Summary

- The basic particle-filter algorithm is now derived!
- After all the detailed math, the method is incredibly simple:
 - For each particle, randomly choose a possible next state of the model.
 - Compute a weight that represents how likely that state happens to be.
- In the rest of this week, we will first implement an example to illustrate the SIS algorithm and will find that it does not perform as well as we might hope.
- Then, our focus shifts to analyzing and fixing the problem with SIS to make the particle filter robust.